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TEI0023 - Communication Interface and Commands

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The modules TEI0015, TEI0016 and TEI0023 implement a handler for executing commands. The serial interface speed must be set to 115200 bits, commands consists of a single character in UTF-8 encoding. Each command must be transmitted individually.

All commands are identical for all tree modules, except commands for setting the gain.

The modules **TEI0015** and **TEI0016** recognizes the following gain commands since module revision 02:

- "**1**" Sets the pre-amplification of the ADC's input to 1
- "**2**" Sets the pre-amplification of the ADC's input to 2
- "**4**" Sets the pre-amplification of the ADC's input to 4
- "**8**" Sets the pre-amplification of the ADC's input to 8

The following gain commands are recognized by all **TEI0023** modules in every module revision:

- "**0**" Deactivates the pre-amplifier
- "**1**" Sets the pre-amplification of the ADC's input to 0.25
- "**2**" Sets the pre-amplification of the ADC's input to 0.5
- "**3**" Sets the pre-amplification of the ADC's input to 1
- "**4**" Sets the pre-amplification of the ADC's input to 2
- "**5**" Sets the pre-amplification of the ADC's input to 4
- "**6**" Sets the pre-amplification of the ADC's input to 8
- "**8**" Sets the pre-amplification of the ADC's input to 16

The modules **TEI0015** and **TEI0023** have an ADC with additional features since revision 02 for **TEI0015** and revision 03 for **TEI0023**:

- "**S**" Activates the Input Span Compression
- "**s**" Deactivates the Input Span Compression
- "**H**" Activates the High-Z Mode
- "**h**" Deactivates the High-Z Mode

These commands are recognized by **every module**:

- "**t**" The ADC measures 1 mega samples and saves the values into its SD-RAM
- "**x**" Instead of ADC values, the value "12345" is stored 1M times into its SD-RAM, values are transmitted via ".", "+" and "*"
- "**y**" Instead of ADC value, hexadecimal values, in ascending order, are generated and stored into the SD-RAM, the values are transmitted in via ".", "+" and "*"

- "**z**" The value "12345" is generated and direct transmitted 256 times
- "**r**" The ADC measure once and transmits this value

- ". " A single value of stored ADC or generated measurement is transmitted
- "+" 128 values of stored ADC or generated measurements are transmitted
- "*" 16 kbit values of stored ADC or generated measurements are transmitted

- "?" The module returns its ID:
TEI0015 with ADC AD4003 / 2 MSps returns "1"
TEI0016-0x-08-C8**A** with ADC ADAQ7988 / 0.5 MSps returns "2"
TEI0016-0x-08-C8**B** with ADC ADAQ7980 / 1 MSps returns "3"
TEI0023A with ADC AD4003 / 2 MSps returns "4"

- "**F**" The module activates a square wave signal, frequency = 10 kHz and amplitude is +3,3 V / ground the signal is accessible on the pads
- D5 in normal mode and
- D6 in time inverted mode
- "**f**" Deactivation of the square wave signal